

SARSHIJ KARN

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Summary

Insightful and dedicated Electronics, Communication & Information Engineering student proficient in electronics, programming, AI, and cybersecurity. Fluent in AI tools and consistently exploring emerging tech trends to sharpen analytical and problem-solving skills. Full-stack developer with hands-on experience building real-world projects in machine learning, computer vision, and web development.

Education

Bachelor's in Electronics, Communication and Information Engineering (BEI) 2024 – Present
National College of Engineering, Tribhuvan University Lalitpur, Nepal

- Pursuing engineering degree focused on electronics, communication systems, and information technology.
- **Full Merit Scholarship** recipient for outstanding academic performance.

Grade XII (NEB) – Science 2023
Liverpool International College Kathmandu, Nepal

- **GPA: 3.53** – Completed higher secondary education with merit-based scholarship.

Grade X (NEB) 2021
Happyland Secondary School Kathmandu, Nepal

- **GPA: 4.00** – Achieved perfect academic performance in secondary education.

Skills

Programming Languages: Python, C/C++, JavaScript

Web & Frontend: HTML5, CSS3, JavaScript, React, TypeScript, Vite, Responsive UI, Frontend Optimization, Project Deployment

AI & Machine Learning: Machine Learning, Deep Learning, TensorFlow, Keras, OpenCV, MediaPipe, BERT, Prompt Engineering, AI Agents

Electronics: Analog & Digital Circuits, Signals & Systems, Circuit Simulation, Arduino & MCUs, PCB Design

Cybersecurity: Linux Security, Ethical Hacking, Terminal Operations, Security Tools, Automation for Security

Tools & Platforms: VS Code, Git, GitHub, GitLab, PowerShell Scripting, Task Automation, Domain & Hosting, Vercel

Soft Skills: Problem Solving, Analytical Thinking, Teamwork, Communication, Time Management, Continuous Learning

Projects

Personal Portfolio Website | *HTML5, CSS3, JavaScript, Vite, GSAP, Vercel* sarshijkarn.com.np

- Engineered a high-performance, fully responsive portfolio leveraging HTML5, CSS3, JavaScript, and Vite, featuring WebGL 3D backgrounds and GSAP animations.
- Implemented advanced SEO techniques (structured data, Open Graph, Twitter Cards) to optimize search visibility and enhance web presence.

BERT-Powered Resume Screener | *Python, BERT, Decision Tree, NLP* GitHub

- Engineered an automated NLP pipeline using BERT to extract and parse key candidate data, significantly reducing manual screening time.
- Implemented a Decision Tree classification model to validate resume authenticity and flag potentially fraudulent applications.

Faceshield – Privacy-First Face Blurring Tool | *Python, MediaPipe, DeepSORT, Flask* GitHub

- Built a robust, privacy-centric video processing application utilizing Python and Flask to automate facial obfuscation.
- Integrated MediaPipe for accurate facial detection and DeepSORT for consistent, real-time multi-object tracking across video frames.

SmartVault – Text and Photo Notes App | *React, TypeScript, Vite, Google Gemini API* GitHub

- Developed a secure, full-stack note-taking application using React, TypeScript, and Vite, featuring a robust local PIN-protected vault for data privacy.
- Integrated the Google Gemini API to enable advanced AI-powered image analysis and enhancement features directly within the user workflow.

MedAssist – Smart Healthcare Companion | *Python, AI, Healthcare*

 [GitHub](#)

- Architected an intelligent healthcare management platform to streamline medication tracking, appointment scheduling, and electronic medical record organization.
- Embedded AI-driven features to provide personalized health insights and optimize user engagement with their medical data.

Gesture-Based Calculator | *Python, OpenCV, MediaPipe, TTS*

 [GitHub](#)

- Designed an interactive, touchless calculator application utilizing MediaPipe for precise hand gesture recognition and mapping.
- Integrated real-time visual overlays via OpenCV and text-to-speech feedback mechanisms to deliver a seamless augmented reality (AR) user experience.

Reddit Data Scraper | *Python, Reddit API, Data Collection*

 [GitHub](#)

- Engineered a scalable Python scraping utility leveraging the Reddit API to programmatically extract and aggregate large volumes of posts and comments.
- Facilitated advanced data collection workflows by implementing chronological filtering and automated data structuring for downstream analysis.

YouTube Channels Bulk Unsubscriber | *Python, Tkinter, YouTube API v3, OAuth 2.0*

 [GitHub](#)

- Developed a streamlined desktop application using Python and Tkinter to automate bulk channel unsubscriptions via the YouTube Data API v3.
- Implemented secure OAuth 2.0 authentication flows to ensure safe, user-authorized interactions with Google account data.

Awards & Recognition

Full Merit Scholarship | *National College of Engineering, T.U.*

2024

Merit-Based Scholarship | *Liverpool International College*

2021 – 2023

Certificate of Participation | *Multiple Hackathons & Workshops*

Interests

Cybersecurity | Technology & AI | Financial Markets | Web & Tech Exploration | Playing Sports | Reading Books | Hackathons | Human Psychology

Languages

English – Professional working proficiency | **Nepali** – Native | **Hindi** – Professional working proficiency | **Maithili** – Native